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Q1.

Resistance is the property of a conductor to oppose the flow of current through it.

SI unit - Ohm (Ω)

Q3.

A fuse melts and breaks the circuit when current exceeds a safe limit, protecting the circuit. A switch is used to manually open or close a circuit and does not melt.

Q2.

Combination: $\text{CaO} + \text{H}_2\text{O} \rightarrow \text{Ca(OH)}_2$

Decomposition: $2\text{FeSO}_4 \rightarrow \text{Fe}_2\text{O}_3 + \text{SO}_2 + \text{SO}_3$

Displacement: $\text{Zn} + \text{CuSO}_4 \rightarrow \text{ZnSO}_4 + \text{Cu}$

Q6.

(a) Electric power is the rate at which electrical energy is consumed. Unit - Watt.

(b) $P = VI$, and since $V = IR$,

$$P = I \times (IR) = I^2R$$

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Q4.

Mendel's law of dominance states that when two contrasting traits are crossed, only one trait (the dominant one) appears in the first generation, while the other (recessive) trait remains hidden.

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Q7.

(i) Do not touch electrical switches with wet hands.

(ii) Use earthing for appliances with metallic bodies.